

Smart Home Sensor: Quick Reference

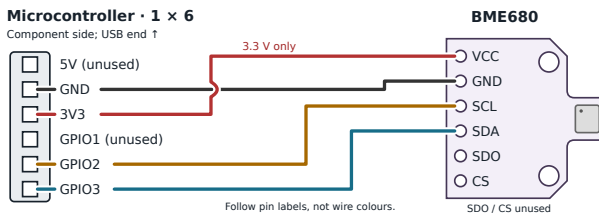
You build a small indoor air-quality and room-climate sensor: air quality, temperature, humidity and pressure on a little display. The sensor can also send its readings to an online dashboard or your own smart-home system (MQTT/Home Assistant compatible).



Build instructions with pictures: scan the QR code

1 Solder and wire the BME680

Thread the stripped ends through the pad holes **from the plain back**, bend them flat against the labelled front and solder there. Then wire the sensor to the board as shown.



Swapped SDA/SCL and wrong board pins are the most common no-sensor causes.

2 Flash and test before assembly

Web flasher: https://alexanderkutschera.com/smart_home_sensor

1. Keep the sensor **outside the enclosure**; plug the board in.
2. Web flasher → select the **workshop image** → **Connect**.
3. Pick the port **USB JTAG/serial debug unit** → **Install**.
4. After the reboot, wait for green **OK** under **BME680 sensor** on the display.

3 Assemble the enclosure

- **Main housing** (bottom)
- **Board:** display facing the open front
- **Lower inlay:** between board and sensor
- **Upper inlay:** BME680 in the recess, **component side up** towards the lid grille, wires out through the side opening
- **Lid** (top)
- **Insulation** (optional) between board and sensor e.g. the board's styrofoam does the job

The lid goes on at an angle, not straight down: slide its **long tab** into the rails, then lower the other edge until the **short tab** snaps in. Release the short tab first to open. The closed lid covers **RESET** — see step 4.

4 Set up on the device

1. First boot opens a Wi-Fi network named like **SHS-Bouncy-Alpaca** (no password) — use the exact name on the display.
2. Browse to **192.168.4.1** → **Configure WiFi** (2.4 GHz only).
3. **Setup**: give the device a name.
4. *Live readings & connection test* → **Send a test reading**. Green ✓ **201** or **200** means stored.
5. **Finish**.

Back into setup later: press RESET twice quickly. The enclosure covers that button — slide the board out, or power up away from the saved Wi-Fi: with no known network it opens setup itself.

5 Your readings

The QR code on the display links straight to your readings. Missed it, or don't know your ID? Go to **diy-sensor.org/dashboard**, find the name you gave your device during setup and select it. New readings arrive every **5 minutes**; the display refreshes every few seconds.

6 Air quality

- **IAQ accuracy 0** stabilizing
- **1-2** calibrating (takes hours)
- **3** trusted — a drop back to 1 is normal, BSEC is rebuilding its baseline
- **IAQ 0-50** good · **51-100** moderate
- **101-150** light · **151-200** moderate
- **201-300** heavy · **300+** severe pollution

Temperature reads high — the board warms the sensor. After 20-30 min, *add* the remaining error to the offset in the setup page: $\text{new} = \text{current} + (\text{reported} - \text{real})$. It starts at 5 °C, not 0.

7 Troubleshooting

- **No port in the flasher?** Use a USB-C **data** cable. Still nothing: unplug, hold **BOOT**, plug in, release after ~2 s.
- **BME680 NOT FOUND:** power off and re-check the four wires — 3V3 (not 5V), GND, SDA → GPIO3, SCL → GPIO2 — and the solder joints.
- **No setup Wi-Fi:** look for the exact name on the display; 2.4 GHz only.
- **Test reading fails:** check the Wi-Fi password and internet access.
- **Nothing on the dashboard yet:** the first reading takes up to 5 min.